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Experiment Name : **TIME SERIES ANALYSIS OF WEBSITE TRAFFIC**

```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose

data = pd.read_csv('website_traffic.csv', parse_dates=['Date'],
index_col='Date')

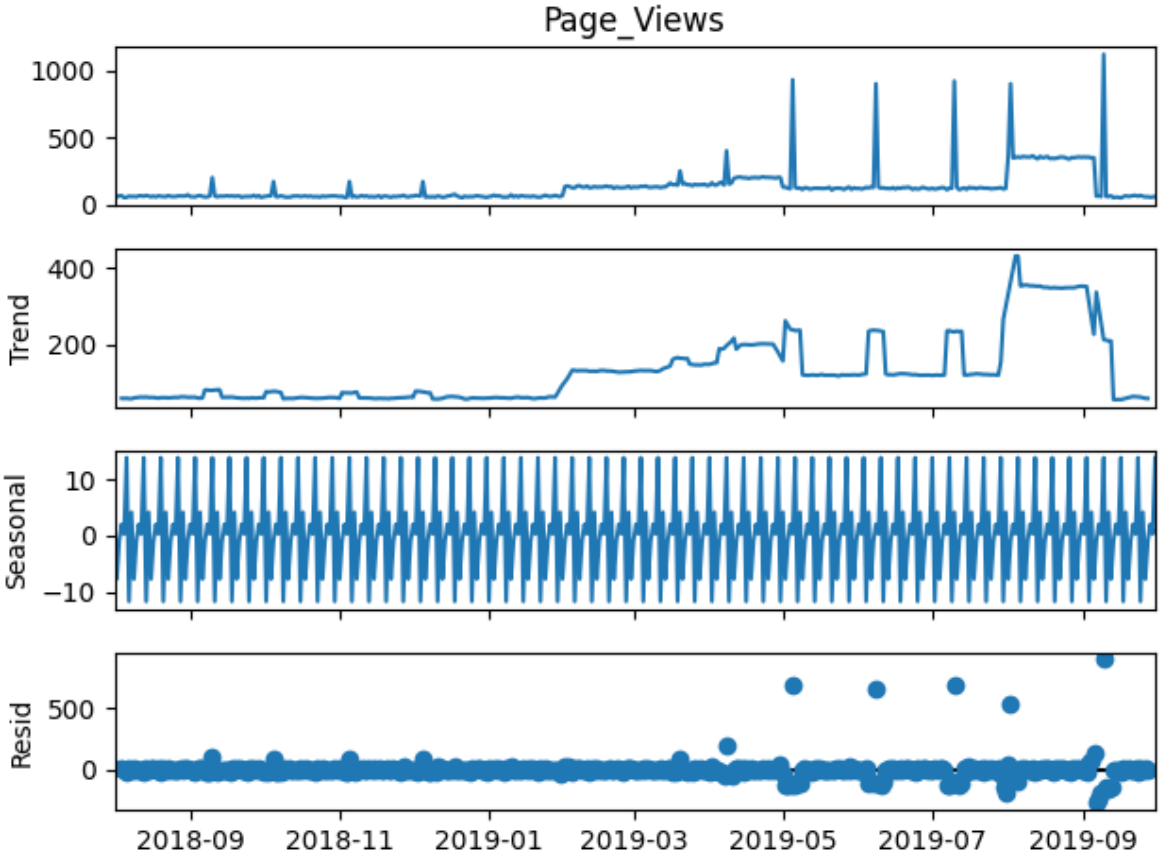
decomposition = seasonal_decompose(data['Page_Views'],
model='additive')
decomposition.plot()
plt.show()

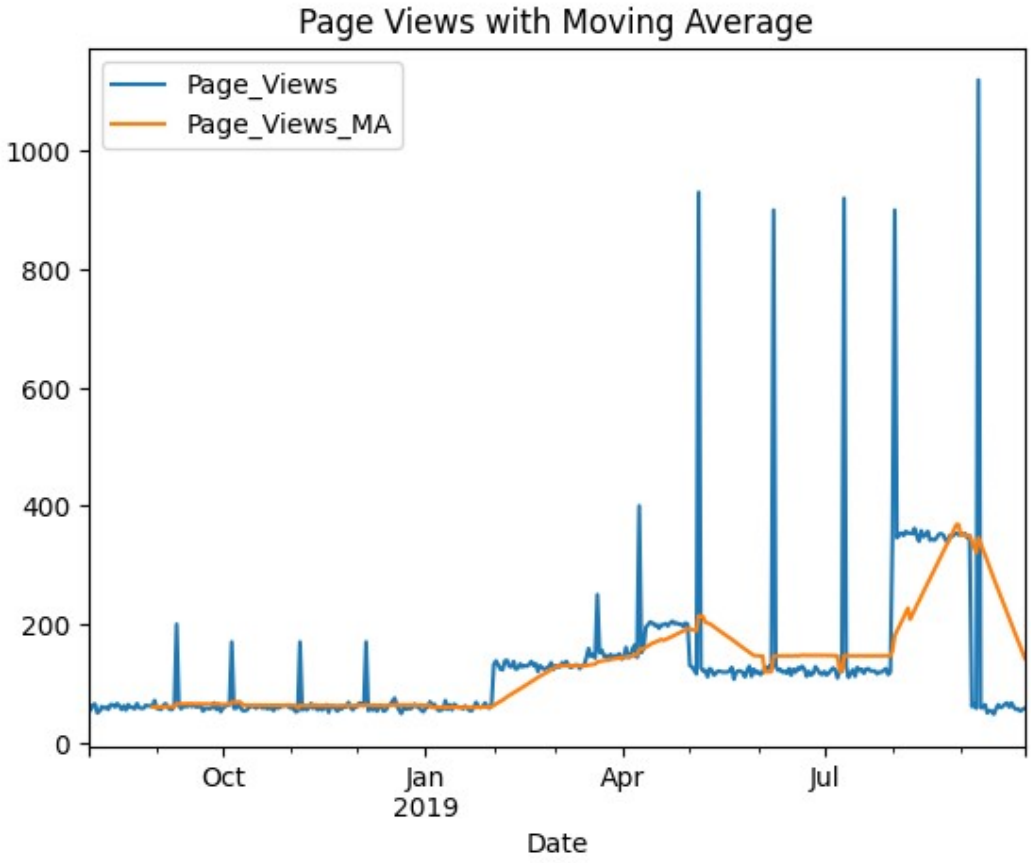
data['Page_Views_MA'] = data['Page_Views'].rolling(window=30).mean()
data[['Page_Views', 'Page_Views_MA']].plot()
plt.title('Page Views with Moving Average')
plt.show()

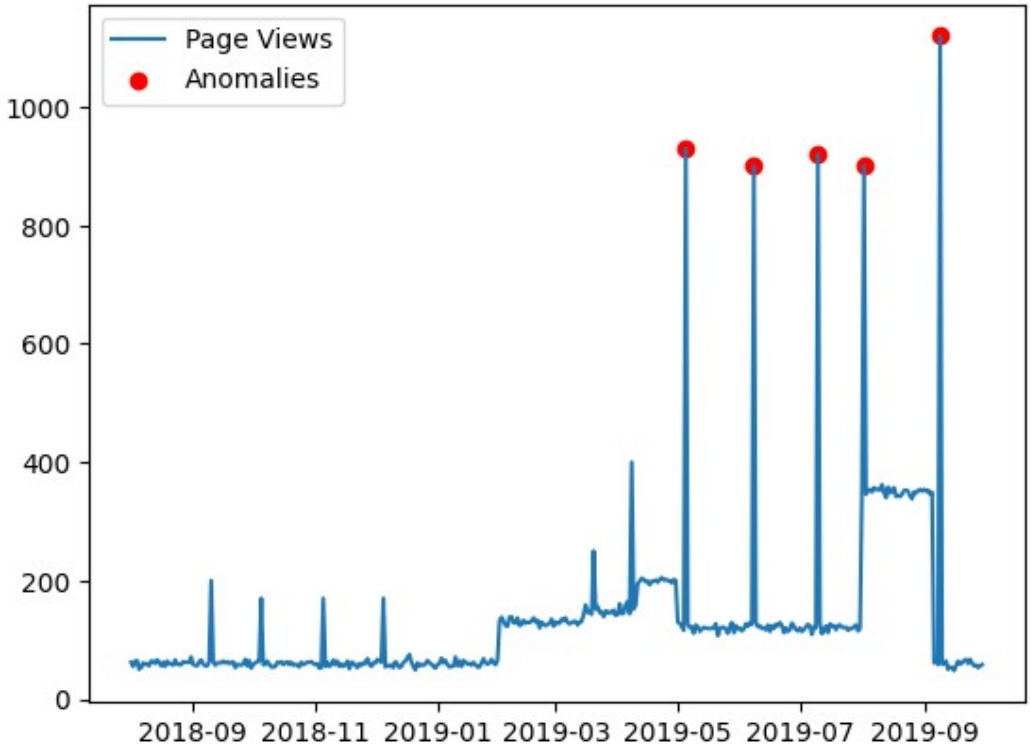
data['Page_Views_Z'] = (data['Page_Views'] -
data['Page_Views'].mean()) / data['Page_Views'].std()

anomalies = data[data['Page_Views_Z'].abs() > 3]

plt.plot(data.index, data['Page_Views'], label='Page Views')
plt.scatter(anomalies.index, anomalies['Page_Views'], color='red',
label='Anomalies')
plt.legend()
plt.show()
```


Page_Views
1000
500
0
Trend
400
200
Seasonal
10
0
−10
Resid
500
0
2018-09
2018-11
2019-01
2019-03
2019-05
2019-07
2019-09


Page Views with Moving Average
Page_Views
Page_Views_MA
1000
800
600
400
200
0
Oct
Jan
2019
Apr
Jul
Date


Page Views
Anomalies
1000
800
600
400
200
0
2018-09
2018-11
2019-01
2019-03
2019-05
2019-07
2019-09